

CHARTERS CREEK, South Africa

WMO: 684910

Lat: 28.1978S

Lon: 32.4142E

Elev: 26

StdP: 14.68

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.1	50.8	38.2	33.9	60.1	43.7	42.0	57.9	16.0	68.6	14.2	71.4	1.7	260	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.2	89.1	76.9	87.1	76.3	85.5	75.7	80.2	85.0	78.9	84.3	77.7	83.1	11.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	151.1	82.5	77.3	142.3	81.7	76.0	136.2	81.0	43.9	85.5	42.4	84.4	41.2	83.2	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.3	14.4	13.2	DB	42.0	96.8	2.6	2.8	40.2	98.9	38.7	100.5	37.3	102.1	35.4	104.1
			WB	40.4	81.7	2.9	1.5	38.4	82.8	36.7	83.7	35.1	84.5	33.0	85.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	71.0	76.6	77.0	76.4	72.4	69.0	65.0	64.3	66.5	68.5	69.7	72.3	75.1	(6)
(7)		DBStd	6.06	3.73	3.59	3.69	4.34	3.74	3.90	4.05	4.55	4.76	4.78	4.58	4.39	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	195	0	0	0	3	9	47	61	36	20	14	4	1	(9)
(10)		CDD50	7678	826	756	818	671	588	451	444	511	555	610	669	779	(10)
(11)		CDD65	2399	361	336	353	224	133	48	40	82	125	159	223	315	(11)
(12)		CDH74	13891	2715	2470	2378	1055	461	167	142	244	465	631	1086	2077	(12)
(13)		CDH80	3081	715	621	572	183	41	14	14	35	88	101	192	505	(13)
(14)	Wind	WSAvg	6.2	6.3	6.3	6.1	5.6	5.2	4.9	5.2	6.4	7.2	7.3	7.1	6.6	(14)
(15)	Precipitation	PrecAvg	35.4	4.6	4.4	4.3	2.6	1.8	1.3	1.3	1.3	2.0	4.2	4.0	3.6	(15)
(16)		PrecMax	61.7	10.6	11.1	10.3	6.8	12.4	5.4	4.8	5.9	8.0	12.8	13.1	8.2	(16)
(17)		PrecMin	16.6	0.6	0.9	0.7	0.6	0.0	0.0	0.0	0.0	0.1	0.9	0.8	1.1	(17)
(18)		PrecStd	10.2	2.6	2.3	2.6	1.5	2.4	1.5	1.3	1.3	1.8	2.8	2.9	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.5	91.3	90.1	87.4	83.7	81.6	82.0	83.9	87.3	87.0	88.1	91.4	(19)
(20)			MCWB	77.5	77.7	78.5	75.1	70.2	65.1	64.2	66.0	70.2	73.8	76.3	78.2	(20)
(21)		2%	DB	87.8	87.8	87.2	84.2	81.0	78.2	77.5	79.1	81.7	82.4	84.4	87.4	(21)
(22)			MCWB	76.5	77.3	77.7	74.4	70.2	65.6	65.0	67.4	69.9	71.8	74.3	77.0	(22)
(23)		5%	DB	85.9	85.7	85.3	81.9	78.9	75.6	74.9	76.3	78.6	79.8	81.9	85.0	(23)
(24)			MCWB	75.6	76.7	76.8	73.4	69.2	65.2	64.7	66.6	68.6	70.7	73.3	75.9	(24)
(25)		10%	DB	84.1	83.9	83.3	79.7	76.6	73.1	72.4	73.9	76.0	77.5	79.7	82.6	(25)
(26)			MCWB	75.1	76.0	75.8	72.2	68.4	64.7	64.0	65.4	67.5	69.5	72.3	74.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.3	82.2	82.1	78.0	73.5	71.1	70.1	72.0	73.1	75.5	77.7	80.1	(27)
(28)			MCDWB	85.3	85.4	85.5	82.7	80.3	75.9	74.8	78.4	83.3	84.1	85.1	87.6	(28)
(29)		2%	WB	78.9	79.9	80.1	76.0	71.7	68.9	67.9	69.5	70.9	73.3	76.0	78.6	(29)
(30)			MCDWB	84.8	84.2	84.5	81.6	78.2	74.2	73.4	75.5	79.7	80.3	82.3	85.6	(30)
(31)		5%	WB	77.5	78.5	78.3	74.4	70.5	67.3	66.3	68.0	69.6	71.8	74.6	77.1	(31)
(32)			MCDWB	83.3	83.6	82.9	80.0	76.5	72.7	72.1	73.9	77.2	77.9	80.2	83.1	(32)
(33)		10%	WB	76.4	77.3	76.7	73.1	69.3	65.8	64.9	66.6	68.4	70.4	73.3	75.6	(33)
(34)			MCDWB	82.0	82.4	81.6	78.3	74.8	71.0	70.6	72.3	74.5	76.1	78.5	81.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.1	13.2	13.8	14.5	16.9	19.4	18.8	16.5	15.2	12.8	12.2	13.0	(35)
(36)			MCDBR	16.2	15.5	15.8	16.3	18.9	21.9	22.0	19.2	18.5	16.3	15.7	16.5	(36)
(37)			MCWBR	7.2	7.1	7.6	7.9	9.6	11.5	11.1	9.5	9.2	8.5	7.9	7.9	(37)
(38)		5% WB	MCDWB	14.3	13.9	14.6	14.9	16.9	18.6	18.6	16.7	17.3	15.1	13.8	15.3	(38)
(39)			MCWBR	7.8	7.2	8.1	7.9	9.7	11.7	11.0	9.2	9.0	8.3	7.7	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.446	0.430	0.430	0.411	0.384	0.371	0.376	0.441	0.493	0.513	0.479	0.465	(40)	
(41)		taud	2.339	2.389	2.377	2.394	2.435	2.448	2.428	2.223	2.096	2.079	2.210	2.267	(41)	
(42)		Ebn at Noon	285	285	274	263	256	252	256	249	251	258	274	281	(42)	
(43)		Edh at Noon	43	40	38	34	30	28	30	40	50	54	49	46	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1941	1892	1649	1328	1149	1009	1075	1265	1442	1558	1753	1921	(44)	
(45)		RadStd	169	141	107	78	50	46	62	77	84	123	156	143	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page